Hydroa Vacciniforme

▲ FIGURE 90-5 Hydroa vacciniforme. Vesicular, bullous, and crusted facial lesions, which are precursors of vacciniform scars.

Lesions typically begin with erythema and sometimes swelling, progressing within 24 hours to symmetrically distributed, often tender papules. These evolve into vesicles, which may be confluent or hemorrhagic, followed by umbilication, crusting, and eventual detachment of the lesions. This process results in permanent, depressed, hypopigmented pock scars within weeks. The rash primarily affects the cheeks, and less commonly other facial areas, the backs of the hands, and the outer aspects of the arms—usually in a symmetric distribution. Rarely, other sites may be involved.

Oral ulcers and ocular manifestations are uncommon but may occur.

Cutaneous Lesions

Hydroa vacciniforme (HV) is characterized by: - Initial erythema ± edema - Development of tender papules within 24 hours - Vesiculation (sometimes hemorrhagic or confluent) - Umbilication - Crusting and lesion detachment - Formation of permanent, atrophic, hypopigmented scars

▲ FIGURE 90-6 Typical hydroa vacciniforme lesions provoked by repeated ultraviolet A irradiation.

Laboratory Tests

Histology

Early histologic findings include intraepidermal vesicle formation, focal keratinocyte necrosis, and spongiosis, accompanied by a dermal perivascular infiltrate of neutrophils and lymphocytes. In older lesions, necrosis, ulceration, and scarring are evident. Vasculitic changes have been reported in rare cases.

Blood Tests
- Assess blood, urine, and stool porphyrin levels to exclude cutaneous porphyria
- Check antinuclear antibody (ANA) and extractable nuclear antigen (ENA) titers to rule out cutaneous lupus erythematosus

Phototesting
- May reveal decreased minimal erythema dose to short-wavelength UVA
- Simulated solar radiation can induce erythema at lower doses or, occasionally, reproduce the characteristic vesicular lesions of HV (see Fig. 90-6)
- However, phototesting is not always specific for HV and may not reliably differentiate it from other photodermatoses

Other Tests
Viral studies (e.g., for herpes simplex virus) should be considered if there is suspicion of a photo-exacerbated viral dermatosis.

Differential Diagnosis

See Box 90-3.

Complications

Pock scarring is an inevitable outcome of HV.

Prognosis and Clinical Course

HV often resolves spontaneously during adolescence, although it may persist into adulthood in some individuals.

Prevention

- Strict sun avoidance
- Use of high-protection-factor, broad-spectrum sunscreens
- Prophylactic phototherapy (e.g., narrowband UVB) administered annually in spring may prevent recurrences in some patients

Treatment

(Refer to eFig. 90-6.1 in the online edition)

Management includes: - Minimizing sun exposure
- Regular use of broad-spectrum, high-SPF sunscreens

Adjunctive therapies with limited or unproven efficacy: - **Antimalarials**: occasional reports of benefit, but efficacy not established
- **β-carotene**: reported anecdotally, but ineffective in some cases (e.g., three cases in our experience)

As with polymorphic light eruption (PMLE), prophylactic phototherapy using narrowband UVB or other regimens may induce tolerance and reduce disease activity.

Box 90-3

Differential Diagnosis of Hydroa Vacciniforme

- Photo-exacerbated viral dermatoses (e.g., herpes simplex)
- Erythropoietic protoporphyria
- Polymorphic light eruption
- Subacute cutaneous lupus erythematosus
- Xeroderma pigmentosum

Chronic Actinic Dermatitis

At a Glance

- A rare, acquired, persistent eczematous eruption affecting sun-exposed skin
- May exhibit pseudolymphomatous features
- Predominantly affects older men; less commonly seen in young atopic individuals
- Rarely associated with hydroa vacciniforme or HIV infection

- Histopathology shows eczematous changes; pseudolymphomatous forms can closely mimic cutaneous T-cell lymphoma